700 Arch diagram

Incus - Simulated cloud environment
tenant-a
web-a
spool
api-a
spool
db-a
spool
bastion-a
spool
tenant-b
web-b
spool
api-b
spool
db-b
spool
bastion-b
spool
tenant-c
web-c
spool
api-c
spool
db-c
spool
bastion-c
spool
tenant-d
web-d
spool
api-d
spool
db-d
spool
bastion-d
spool
Incus host
audit.d log
EvidenceSpool batches
Acquisition Layer
Instance EvidenceAgent
Agent configuration
Collection scheduler / run loop
Collection subsystem
InstanceEventCollector
System / process activity collector
Filesystem or audit-related collector
Evidence normalisation
Attach acquisition context
Evidence identity and integrity Deterministic ID/ SHA-256
Evidence batch / delivery payload
Persistence local spool
Central API delivery
Delivery successful?
Yes
No
Remove delivered spool item
Retain pending item and rety on later loop
HostEvidenceAgent
Incremental Host Audit Collector
IncusAuditAttributor
EvidenceEnvelope
EvidenceBatch
POST /api/v1/evidence/batches
Persist checkpoint
Retry / Delivery
POST /api/v1/evidence/events
Tenant Resolver
Evidence Packaging
Acquisition Metadata
Preservation Layer
Evidence Ingestion API
Request and evidence correlation
Evidence Normalisation
Integrity Verification and Hash calculation (SHA256)
Provenance and Chain-of-Custody Metadata
Capture Manifest
Transactional Evidence Persistence
Central Evidence Storage
Evidence Integrity and Provenance records
Central PostgreSQL DB
Read Only
Analysis Layer
Correlation Engine
PostgreSQL Correlation Source
EvidenceNormaliser host and instance adapters
Controls
Evidence integrity SHA-256 and provenance references
Privacy controls
Tenant context
Explainability
Feature values
Weights
Contributions
Threshold
CorrelationContext
Canonical Evidence identity, tenant, event time, collector, actor, resource, raw bytes, SHA-256
Unique evidence IDs within analysis?
no
DuplicateID error - Anaysis fails
yes
AttributeExtractor canonical fields + source attributes
EntityExtractor tenant, project, instance, user, resource, process, stable IDs
RelationshipMapper belongs_to, acted_on, occurred_
CandidateIndex temporal teimstamps index + entity
Unique evidence pairs
CandidateGenerator tenant IDs, shared entities/resources, actions, evidence references, provenance
Deterministic CorrelationCandidate
Scope and policy gates
Candidate belongs to this analysis
Yes
No
CrossTenantPolicy
no
Policy/Scope denied Score and findings not created
denied
Record policy exclusion
allowed
CorrelationScorer Default: temporal=0.25; entity=0.25; resource=0.20; action=0.15; provenance=0.15
Feature values, weights, contributions, evidence attributes used and explanations permitted features are renormalised
Score >= context minimum score?
Yes
No
Causal reasoning
Reject candidate; rejection count increments
Resolve supported actions and establish trustworthy event order
Check temporal window and, when required, shared-resource compatibility
Zero or more explainable links
Create one of the lines: REQUIRES / PROVIDES / ENABLES with confidence and rationale
No supported causal link
Build CorrelationFinding score + explanation + policy decision + source references + SHA-256 provenance + causal assessment
EvidenceGraphBuilder accepts final policy-approved finding
Deterministic finding de-duplication
Nodes: events, tenant, projects, entites/respurces and findings
Edges: occurred-in, entity relationship, supports-finding, correlated-with
CorrelationResult findings + evidence graph + counts + duration +candidate reduction + provenance completeness + warnings
Presenation Layer
Dashboard service
Dashboard
Overview Page
Tenant Page
Evidence Page
Evidence Detail Page
Correlation Page